OptiSense Triage – Project Summary

"Transforming Emergency Care with AI — delivering faster, more accurate triage decisions, better patient outcomes, and smarter resource allocation. Together, we can revolutionize emergency department efficiency and save lives."

Overview

OptiSense Triage is a real-time AI decision support system for emergency department nurses. It delivers a calibrated severity score — aligned with the Emergency Severity Index (ESI Level 1–5) — within seconds of patient arrival, alongside a plain-English clinical summary, the top three danger signals, and an ordered action plan. The system is designed not to replace the nurse's judgment, but to give them a reliable, data-grounded second opinion at the moment it matters most.

Innovation Approach

Manual triage in busy emergency departments carries a documented misclassification rate of 30–40%. Nurses assess dozens of patients simultaneously under extreme cognitive load, with no algorithmic support. OptiSense Triage attacks this problem at the point of care using a three-layer AI architecture:

Layer 1 — Predictive Model: An XGBoost classifier trained on real ICU admissions from the MIMIC-III Demo clinical database. The model ingests standard vitals (heart rate, SpO_2, temperature, respiratory rate, Blood Pressure), a chief complaint encoded via sentence-transformer embeddings, and pre-existing comorbidities — producing a calibrated critical vs. non-critical decision in under a few seconds.

Layer 2 — Clinical Safety Override: Rather than trusting the model blindly, OptiSense Triage applies a rule-based clinical override engine on top of every prediction. Hard vital-sign thresholds — drawn from ACEP ESI guidelines — force immediate escalation when a patient presents with critical hypotension, severe hypoxia, extreme tachycardia, or respiratory failure risk, regardless of what the model predicted. This layer is the safety floor: the AI cannot under-triage a patient whose vitals are objectively dangerous.

Layer 3 — LLM Clinical Reasoning: The model output and clinical signal engine findings are passed to Gemini 2.5 Flash, which generates a charge-nurse-ready summary: a one-sentence risk profile, the top three clinically dangerous findings explained in plain English, and 3–5 ordered next steps appropriate for the assigned ESI level. If Gemini is unavailable, a rule-based fallback generates equivalent structured output — the system degrades effeciently without losing clinical safety.

The full decision is surfaced through a React + Tailwind CSS dashboard that includes a live severity-sorted patient queue, bed availability by category, and escalation flags for cases requiring mandatory human review — giving the charge nurse a real-time risk profile of the entire department at a glance.

Unique Differentiators

1. **A Safety Layer the Model Cannot Override**
   Most clinical AI tools produce a prediction and stop. OptiSense Triage treats the model as a prior, not a final answer. The clinical override engine and human-review flag operate independently of the model — meaning a dangerous patient is escalated even if the model is wrong. This architecture is a meaningful step toward regulatory-ready clinical AI.
2. **Trained on Real Clinical Data, Not Synthetic Benchmarks**
   The MIMIC-III dataset (MIT/Beth Israel Deaconess Medical Center) contains de-identified records from real ICU admissions. Training on this data grounds OptiSense Triage's predictions in actual clinical outcomes.
3. **Free-Text Complaint Understanding**
   Unlike systems that require structured complaint codes, OptiSense Triage accepts a plain English chief complaint from the nurse, encodes it using a sentence-transformer model (all-MiniLM-L6-v2), and uses it as a model feature. The same text is passed to Gemini for clinical context. This bridges the gap between how nurses communicate and how models compute.
4. **Department-Level Situational Awareness**
   Beyond individual patient scoring, the dashboard aggregates live severity data across all active patients — with a severity-sorted queue, bed utilisation by category, and escalation flags — giving charge nurses a continuous, prioritised view of departmental risk that does not exist in current EMR workflows.

Real-World Implementation Potential

OptiSense Triage is architected for clinical integration from day one:

- **EMR Integration:** The backend is a FastAPI microservice exposing a standard REST API with four endpoints. Integration with existing Electronic Health Records (EHRs) would definitely be possible.
- **HIPAA-Ready by Design:** The inference pipeline is stateless. No patient data is persisted at the application layer beyond the in-session queue. All data in transit is encrypted.
- **Regulatory Pathway:** OptiSense Triage is positioned as a Software as a Medical Device (SaMD) decision support tool under FDA guidelines. The explainability architecture — including the override layer audit trail, human-review flags, and Gemini-generated confidence explanations — directly supports the documentation requirements for AI/ML-based SaMD submissions.
- **Scalability:** Cloud-native deployment on Google Cloud Run enables per-site SaaS licensing. The model architecture makes retraining on new hospital data fast and computationally inexpensive. Model accuracy improves continuously as outcome data accumulates.

Target Market: US emergency departments (140M+ annual visits, CDC), expanding to EU and MENA healthcare systems. Initial go-to-market targets high-volume urban ERs and trauma centres where misclassification risk and resource pressure are highest.